Mathematics for ML: Vectors & Vector Spaces

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Outline

1 Vectors & Vector Spaces

About Me

Yogesh Haribhau Kulkarni

Bio:

- 20+ years in CAD/Engineering software development
- Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
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About Content

- Bare essential mathematics needed for Machine/Deep Learning
- Just to get you started.
- Each field within Machine/Deep Learning can go extremely deep.
- This is to give you enough foundation needed.

About Me

- I am not a mathematician.
- And this course won't turn YOU into one!!! (if you are not already)
- But, I will surely attempt to make you get interested in the learning, more.

Linear Algebra

What's in a name?

- "Algebra" means, roughly, "relationship", between unknown numbers.
- Without knowing x and y, we can still work out that $(x+y)^2 = x^2 + 2xy + y^2$
- "Linear Algebra" means, roughly, "line-like relationships".
- Meaning, not curve like, ie quadratic, cubic, sinusoidal, etc, right?
- Nothing in "power" term!!
- An operation F is linear if scaling inputs scales the output, and adding inputs adds the outputs:

$$F(ax) = a.F(x)$$
$$F(x+y) = F(x) + F(y)$$

- When plotted, its a line!!

(Ref: An Intuitive Guide to Linear Algebra - Better Explained)

Linear Equations

- $F(x, y, z) = 3x + 4y + 5z$
- Whats an example for this?
- Can you represent this by multiplication of two vectors?

(Ref: An Intuitive Guide to Linear Algebra - Better Explained)

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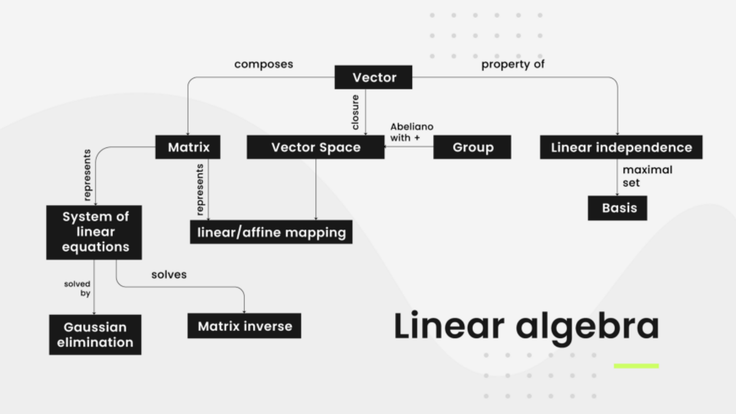
Basic Entities

- Scalars?
- Vectors?
- Matrices?
- Next? (or Whats this called collectively?)
- Point in n-dimensional space is represented by?

Landscape: Linear Algebra



(Ref: The NOT definitive guide to learning math for machine learning - Favio Vazquez)

Vectors

Vectors

- At its simplest, a vector is an entity that has both magnitude and direction.
- The magnitude represents a distance (for example, “2 miles”) and the direction indicates which way the vector is headed (for example,“East”).
- One more way is $\bar{v} = 2\hat{i} + 3\hat{j}$; meaning?
- Is Magnitude-Direction form equivalent to i-j form?
- Inter-convertible? How?
- Can it have just two components?

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Vectors

Two-dimensional example:

- A vector that is defined by a point in a two-dimensional plane
- A two dimensional coordinate consists of an x and a y value, and in this case we'll use 2 for x and 1 for y
- Its is written in matrix form as : $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
- Describes the movements required to get to the end point (of head) of the vector
- So, it is not a point in space. It gives Direction, like a movement recipe.
- When added to a point, results into a transformed point.
- In this case, we need to move 2 units in the x dimension, and 1 unit in the y dimension

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Vectors

Two-dimensional example:

- Note that we don't specify a starting point for the vector
- We're simply describing a destination coordinate that encapsulate the magnitude and direction of the vector.
- Think about it as the directions you need to follow to get to **there** from **here**, without specifying where **here** actually is!
- Generally using the point 0,0 as the starting point (or origin). Also called as Position Vector.
- Our vector of (2,1) is shown as an arrow that starts at 0,0 and moves 2 units along the x axis (to the right) and 1 unit along the y axis (up).

Vectors

Calculating Magnitude

- $\|\vec{v}\| = \sqrt{{v_1}^2 + {v_2}^2}$
- Double-bars are often used to avoid confusion with absolute values.
- Note that the components of the vector are indicated by subscript indices $(v_1, v_2, \ldots v_n)$
- In this case, the vector v has two components with values 2 and 1, so our magnitude calculation is:
- $\|\vec{v}\| = \sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.24$

Vectors

Calculating Direction

- We can get the angle of the vector by calculating the inverse tangent; sometimes known as the arctan
- For our v vector (2,1):$tan(\theta) = \frac{1}{2}$
- $\theta = tan^{-1}(0.5) \approx 26.57^o$
- use the following rules:
  - Both x and y are positive: Use the tan-1 value.
  - x is negative, y is positive: Add 180 to the tan-1 value.
  - Both x and y are negative: Add 180 to the tan-1 value.
  - x is positive, y is negative: Add 360 to the tan-1 value.

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Vectors

- Vectors are defined by an n-dimensional coordinate that describe a point in space that can be connected by a line from an arbitrary origin.
- Are n-dimensional Points and Vectors equivalent? How?
- $\|\vec{v}\| = \sqrt{v_1{}^2 + v_2{}^2 ... + v_n{}^2}$

Definition A *vector* is a matrix with one column.

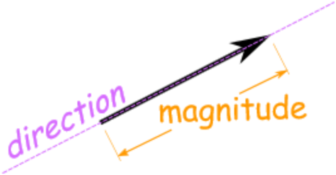
Example

$$\begin{bmatrix} 1 \\ 2 \\ -5 \\ 9 \end{bmatrix}$$

Note Two vectors are equal precisely when they have the same number of rows and all their corresponding entries are equal.

Vectors (Recap)

- A vector has magnitude (how long it is) and direction
- A point can be a vector (position vector, from Origin)
- A data row is a point in n-dimensions, thus a vector as well.



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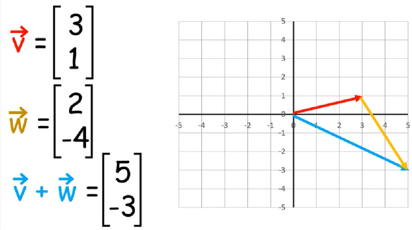
Vector Addition



- To add these vectors: We just add the individual components, so 3 plus 2 is 5; and 1 plus -4 is -3.
- It is simply adding another leg to the journey; so if we follow vector V along 3 and up 1, and then follow vector W along 2 and down 4, we end up at the head of the vector we calculated by adding V and W together.

Vector Addition

Definition We define the sum and of two vectors by

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

and the product of a scalar and a vector by

$$\alpha \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} \alpha u_1 \\ \alpha u_2 \\ \vdots \\ \alpha u_n \end{bmatrix}$$

Example

$$\begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix} \qquad \text{and} \qquad 3\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 3 \end{bmatrix}$$

Exercise

Let $\vec{u}$ and $\vec{v}$ be given by

$$\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \text{and} \qquad \vec{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Plot $\vec{u}$, $\vec{v}$, $\vec{2u}$ and $\vec{u} + \vec{v}$.

Parallelogram rule for vector addition Suppose $\vec{u}$ and $\vec{v} \in \mathbb{R}^2$. Then $\vec{u} + \vec{v}$ corresponds to the fourth vertex of the parallelogram whose opposite vertex is $\vec{0}$ and whose other two vertices are $\vec{u}$ and $\vec{v}$.

Exercise

Let $\vec{u} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$. Display $\vec{u}$, $-2/3\vec{u}$, $\vec{v}$ and $-2/3\vec{u} + \vec{v}$ on a graph.

$\mathbb{R}^n$

In general we will consider vectors in $\mathbb{R}^n$, that is, having n real entries.

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \in \mathbb{R}^n$$

The zero vector is $\vec{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ having n entries, each equal to 0.

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Properties of R^n

Theorem Suppose that $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and $c, d \in \mathbb{R}$. Then,

- $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.
- $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
- $\vec{u} + \vec{0} = \vec{0} + \vec{u} = \vec{u}$
- $\vec{u} + -\vec{u} = -\vec{u} + \vec{u} = \vec{0}$ $\quad (-\vec{u} = (-1)\vec{u})$
- $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$
- $(c + d)\vec{u} = c\vec{u} + d\vec{u}$
- $c(d\vec{u}) = (cd)\vec{u}$
- $1 \cdot \vec{u} = \vec{u}$

Vector Spaces

Background

- Vectors are generally represented by list of numbers, these are coefficients of the basis vectors. For 2D, x and y axes are basis and coefficients are the coordinates.
- We can have n-dimensional vector representing point in n-dimensional space.
- Determinant (area notion) and eigen vectors (principal direction) are independent of the basis vectors ie the coordinate system, ie. they have some spatial ie space-ness properties.
- SImilarly even "functions" behave similarly. Instead of discrete numbers like earlier, they represent continuous numbers. But the space-ness remains.
- The way you can add two vectors, you can add two functions. As you can scale a vector, you can scale a function. The resultant function curves just add up or scale respectively.

(Ref: Abstract vector spaces — Chapter 16, Essence of linear algebra - 3Blue1Brown)

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Background

- Addition and scaling thus is common to vectors and function.
- Linear combination has both addition and scaling. Its a Transformation or Operator.
- Transformation is linear if it satisfies two properties : addition and scaling.
- Scaling of an addition is addition of the scaled vectors.
- Transformation of vectors can also be done by representing it in a transformed coordinate system (ie basis vectors)
- This is true for vectors as well as functions.
- Derivatives also has additive and scaling properties. Meaning, derivative of an addition is addition of the derivatives. Same for scaling.

(Ref: Abstract vector spaces — Chapter 16, Essence of linear algebra - 3Blue1Brown)

Background

Derivative is Linear

$$L(\vec{\mathbf{v}} + \vec{\mathbf{w}}) = L(\vec{\mathbf{v}}) + L(\vec{\mathbf{w}})$$

$$\frac{d}{dx}(x^3 + x^2) = \frac{d}{dx}(x^3) + \frac{d}{dx}(x^2)$$

$$L(c\vec{\mathbf{v}}) = cL(\vec{\mathbf{v}})$$

$$\frac{d}{dx}(4x^3) = 4\frac{d}{dx}(x^3)$$

(Ref: Abstract vector spaces — Chapter 16, Essence of linear algebra - 3Blue1Brown)

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Background

Polynomials can also be represented as coefficients of basis having powers of x.

Our current space: All polynomials

$$5 \cdot 1 + 3x + 1x^2 + 0x^3 + 0x^4 + \cdots = \begin{bmatrix} 5 \\ 3 \\ 1 \\ 0 \\ 0 \\ \vdots \end{bmatrix}$$

Basis functions

$b_0(x) = 1$

$b_1(x) = x$

$b_2(x) = x^2$

$b_3(x) = x^3$

$\vdots$

Vectors and functions have similar options.

Linear algebra concepts	Alternate names when applied to functions
Linear transformations	Linear operators
Dot products	Inner products
Eigenvectors	Eigenfunctions

(Ref: Abstract vector spaces — Chapter 16, Essence of linear algebra - 3Blue1Brown)

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Background

- So many things are Vector-ish, adhering to Learning Transformation.
- Thus all those have been abstracted as a structure called "Vector Spaces". Thus while developing further theory, we do not have to think about actual reincarnations.
- So a set of rules (called 'Axioms') are decided to define the "Vector Spaces".
- Axioms are interfaces or protocols, whoever abides by them can thus be applicable in the further theory or proofs.

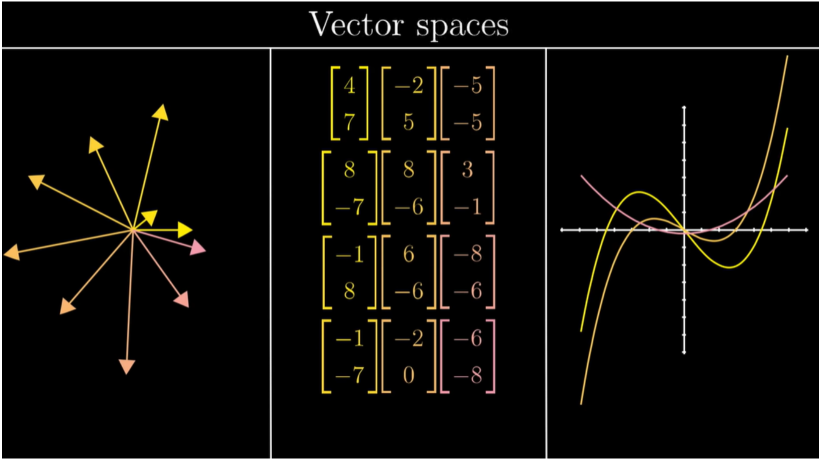
Rules for vectors addition and scaling

1. $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$
2. $\vec{v} + \vec{w} = \vec{w} + \vec{v}$
3. There is a vector $\mathbf{0}$ such that $\mathbf{0} + \vec{v} = \vec{v}$ for all $\vec{v}$
4. For every vector $\vec{v}$ there is a vector $-\vec{v}$ so that $\vec{v} + (-\vec{v}) = \mathbf{0}$
5. $a(b\vec{v}) = (ab)\vec{v}$
6. $1\vec{v} = \vec{v}$
7. $a(\vec{v} + \vec{w}) = a\vec{v} + a\vec{w}$
8. $(a + b)\vec{v} = a\vec{v} + b\vec{v}$

"Axioms"

(Ref: Abstract vector spaces — Chapter 16, Essence of linear algebra - 3Blue1Brown)

Background



(Ref: Abstract vector spaces — Chapter 16, Essence of linear algebra - 3Blue1Brown)

What determines a vector space?

- A nonempty set V whose elements are called *vectors*.
- An operation $+$ called vectors addition, such that

$$\text{vector} \;+\; \text{vector} \;=\; \text{vector}$$

  In other words, we have *closure* under vector addition.
- An operation $\cdot$ called scalar multiplication, such that

$$\text{scalar} \;\cdot\; \text{vector} \;=\; \text{vector}$$

  In other words, we have *closure* under scalar multiplication.
- The remaining 8 axioms are all satisfied.

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Some basic identities in a vector space

Theorem: Let V be a vector space. The following statements are always true.

A) $0 \cdot \vec{u} = \vec{0}$

B) $k \cdot \vec{0} = \vec{0}$

C) $(-1) \cdot \vec{u} = -\vec{u}$

D) If $k \cdot \vec{u} = \vec{0}$ then $k = 0$ or $\vec{u} = \vec{0}$.

Vector subspace

Let $(V, +, \cdot)$ be a vector space.
Definition: A subset W of V is called a *subspace* if W is itself a vector space with the operations $+$ and $\cdot$ defined on V.
Theorem: Let W be a nonempty subset of V. Then W is a subspace of V if and only if it is *closed* under addition and scalar multiplication, in other words, if:

I.) For any $\vec{u}, \vec{v}$ in W, we have $\vec{u} + \vec{v}$ is in W.

II.) For any scalar k and any vector $\vec{u}$ in W we have $k \cdot \vec{u}$ is in W.

Linear combinations and the span of a set of vectors

Let $(V, +, \cdot)$ be a vector space.

Definition: Let $\vec{v}$ be a vector in V. We say that $\vec{v}$ is a *linear combination* of the vectors $\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}$ if there are scalars $k_1, k_2, \ldots, k_r$ such that

$$\vec{v} = k_1 \cdot \vec{v_1} + k_2 \cdot \vec{v_2} + \ldots + k_r \cdot \vec{v_r}$$

The scalars $k_1, k_2, \ldots, k_r$ are called the *coefficients* of the linear combination.

Definition: Let $S = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\}$ be a set of vectors in V.

Let W be the set of **all linear combinations** of $\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}$:

$$W = \{k_1 \cdot \vec{v_1} + k_2 \cdot \vec{v_2} + \ldots + k_r \cdot \vec{v_r} \;\; : \;\text{for all choices of scalars } k_1, k_2, \ldots, k_r\}$$

Then W is called the *span* of the set S.

We write:

$$W = \text{ span } S \quad \text{ or } \quad W = \text{ span } \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\}$$

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Linear combinations and the span of a set of vectors

Let $(V, +, \cdot)$ be a vector space and let $S = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\}$ be a set of vectors in V.

$$\text{span} S = \{k_1 \cdot \vec{v_1} + k_2 \cdot \vec{v_2} + \ldots + k_r \cdot \vec{v_r} \ : \ \text{for all scalars } k_1, k_2, \ldots, k_r\}$$

(all linear combinations of $\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}$).

Theorem: span $\{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\}$ is a subspace of V.
It is in fact the *smallest* subspace of V that contains all vectors $\{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\}$.

Linear independence in a vector space V

Let V be a vector space and let $S = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\}$ be a set of vectors in V.
Definition: The set S is called *linearly independent* if the vector equation

$$(*) \quad c_1 \cdot \vec{v_1} + c_2 \cdot \vec{v_2} + \ldots + c_r \cdot \vec{v_r} = \vec{0}$$

has **only one** solution, the trivial one:

$$c_1 = 0,\ c_2 = 0,\ \ldots, c_r = 0$$

The set is called linearly dependent otherwise, if equation (*) has other solutions besides the trivial one.
Theorem: The set of vectors S is linearly independent if and only if **no** vector in the set is a linear combination of the other vectors in the set.
The set of vectors S is linearly dependent if and only if one of the vectors in the set is a linear combination of the other vectors in the set.

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Introduction

- Thus far we have thought of vectors as lists of numbers in $\mathbb{R}^n$.
- As it turns out, the idea of a vector can be much more general.
- In the spirit of generalization, then, we will define vectors based on their most important properties.
- Once complete, our new definition of vectors will include vectors in $\mathbb{R}^n$, but will also cover many other extremely useful notions of vectors.
- We do this in the hope of creating a mathematical structure applicable to a wide range of real world problems.
- The two key properties of vectors are that they can be added together and multiplied by scalars.

Over to definition . . .
(Ref: Linear Algebra - UCDavis, David Cherney, Tom Denton, Rohit Thomas and Andrew Waldron)

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Definition

Definition

A *vector space* (over $\mathbb{R}$) is a set V with two operations $+$ and $\cdot$ satisfying the following properties for all $u, v \in V$ and $c, d \in \mathbb{R}$:

(+I) (Additive Closure) $u + v \in \mathbb{R}$. (Adding two vectors gives a vector.)

(+II) (Additive Commutativity) $u + v = v + u$. (Order doesn't matter.)

(+III) (Additive Associativity) $(u + v) + w = u + (v + w)$ (Order of adding many vectors doesn't matter.)

(+IV) (Zero) A special vector $0_V \in V$ such that $u + 0_V = u$ for all u in V.

(+V) (Additive Inverse) For every $u \in v$ there exists $w \in V$ such that $u + w = 0_V$.

(Ref: Linear Algebra - UCDavis, David Cherney, Tom Denton, Rohit Thomas and Andrew Waldron)

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Definition

DEFINITION

A *vector space* (over $\mathbb{R}$) is a set V with two operations $+$ and $\cdot$ satisfying the following properties for all $u, v \in V$ and $c, d \in \mathbb{R}$:

($\cdot$ I) (Multiplicative Closure) $c \cdot v \in V$. (Scalar times a vector is a vector.)

($\cdot$ II) (Distributivity) $(c + d) \cdot v = c \cdot v + d \cdot v$. (Scalar multiplication distributes over addition of scalars.)

($\cdot$ III) (Distributivity) $c \cdot (u + v) = c \cdot u + c \cdot v$. (Scalar multiplication distributes over addition of vectors.)

($\cdot$ IV) (Associativity) $(cd) \cdot v = c \cdot (d \cdot v)$.

($\cdot$ V) (Unity) $1 \cdot v = v$ for all $v \in V$.

(Ref: Linear Algebra - UCDavis, David Cherney, Tom Denton, Rohit Thomas and Andrew Waldron)

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Definition

Remark

Don't confuse the scalar product $\cdot$ with the dot product $\cdot$. The scalar product is a function that takes a vector and a number and returns a vector. (In notation, this can be written $\cdot : \mathbb{R} \times V \to V$.) On the other hand, the dot product takes two vectors and returns a number. (In notation: $\cdot : V \times V \to \Re$.)
Once the properties of a vector space have been verified, we'll just write scalar multiplication with juxtaposition $cv = c \cdot v$, though, to avoid confusing the notation.

Remark

It isn't hard to devise strange rules for addition or scalar multiplication that break some or all of the rules listed above.
One can also find many interesting vector spaces, such as the following.

(Ref: Linear Algebra - UCDavis, David Cherney, Tom Denton, Rohit Thomas and Andrew Waldron)

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Example

Example

$$V = \{f | f : \mathbb{N} \to \Re\}$$

Here the vector space is the set of functions that take in a natural number n and return a real number. The addition is just addition of functions: $(f_1 + f_2)(n) = f_1(n) + f_2(n)$. Scalar multiplication is just as simple: $c \cdot f(n) = cf(n)$.

We can think of these functions as infinite column vectors: $f(0)$ is the first entry, $f(1)$ is the second entry, and so on. Then for example the function $f(n) = n^3$ would look like this:

$$f(n) = \begin{pmatrix} 0 \\ 1 \\ 8 \\ 27 \\ \vdots \\ n^3 \end{pmatrix}$$

(Ref: Linear Algebra - UCDavis, David Cherney, Tom Denton, Rohit Thomas and Andrew Waldron)

Example

Example

Alternatively, V is the space of sequences: $f = \{f_1, f_2, \ldots, f_n, \ldots\}$.
Let's check some axioms.

(+I) (Additive Closure) $f_1(n) + f_2(n)$ is indeed a function $\mathbb{N} \to \Re$, since the sum of two real numbers is a real number.

(+IV) (Zero) We need to propose a zero vector. The constant zero function $g(n) = 0$ works because then $f(n) + g(n) = f(n) + 0 = f(n)$.

The other axioms that should be checked come down to properties of the real numbers.

(Ref: Linear Algebra - UCDavis, David Cherney, Tom Denton, Rohit Thomas and Andrew Waldron)

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Example

EXAMPLE

Another very important example of a vector space is the space of all differentiable functions:

$$\{f | f : \Re \rightarrow \Re, \ \frac{d}{dx} f \text{ exists}\}.$$

The addition is point-wise $(f + g)(x) = f(x) + g(x)$, as is scalar multiplication $c \cdot f(x) = cf(x)$.
From calculus, we know that the sum of any two differentiable functions is differentiable, since the derivative distributes over addition. A scalar multiple of a function is also differentiable, since the derivative commutes with scalar multiplication $(\frac{d}{dx}(cf) = c\frac{d}{dx}f)$. The zero function is just the function such that $0(x) = 0$ for every x. The rest of the vector space properties are inherited from addition and scalar multiplication in $\Re$.
In fact, the set of functions with at least k derivatives is always a vector space, as is the space of functions with infinitely many derivatives.

(Ref: Linear Algebra - UCDavis, David Cherney, Tom Denton, Rohit Thomas and Andrew Waldron)

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Example

Remark (Vector Spaces Over Other Fields)

Above, we defined vector spaces over the real numbers. One can actually define vector spaces over any field. A field is a collection of "numbers" satisfying a number of properties.

One other example of a field is the complex numbers,

$$\mathbb{C} = \{x + iy | i^2 = -1, x, y \in \Re\}.$$

In quantum physics, vector spaces over $\mathbb{C}$ describe all possible states a system of particles can have.
For example,

$$V = \{ \begin{pmatrix} \lambda \\ \mu \end{pmatrix} : \lambda, \mu \in \mathbb{C}\}$$

describes states of an electron, where $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ describes spin "up" and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ describes spin "down". Other states, like $\begin{pmatrix} i \\ -i \end{pmatrix}$ are permissible, since the base field is the complex numbers.

Complex Numbers

- Complex numbers have a special property: every polynomial over the complex numbers factors into a product of linear polynomials.
- For example, the polynomial $x^2 + 1$ doesn't factor over the real numbers, but over the complex numbers it factors into $(x + i)(x - i)$.
- Another useful field is the rational numbers $\mathbb{Q}$. A real number given by an infinite string of numbers after the decimal point can't be stored by a computer. So instead rational approximations are used. Since the rationals are a field, the mathematics of vector spaces still apply to this special case.
- One example of field is $\mathbb{Z}_2$ which only has elements $\{0, 1\}$. Multiplication is defined normally, and addition is the usual addition, but with $1 + 1 = 0$. Modern computers actually use $\mathbb{Z}_2$ arithmetic for every operation.
- In fact, for every prime number p, the set $\mathbb{Z}_p = \{0, 1, \ldots, p - 1\}$ forms a field.

(Ref: Linear Algebra - UCDavis, David Cherney, Tom Denton, Rohit Thomas and Andrew Waldron)

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The definition of a vector space $(V, +, \cdot)$

- For any $\vec{u}$ and $\vec{v}$ in V, $\vec{u} + \vec{v}$ is also in V.
- For any $\vec{u}$ and $\vec{v}$ in V, $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.
- For any $\vec{u}$, $\vec{v}$, $\vec{w}$ in V, $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$.
- There is an element in V called the *zero* or *null* vector, which we denote by $\vec{0}$, such that for all $\vec{u}$ in V we have $\vec{0} + \vec{u} = \vec{u}$.
- For every $\vec{u}$ in V, there is a vector called the *negative* of $\vec{u}$ and denoted $-\vec{u}$, such that $-\vec{u} + \vec{u} = \vec{0}$.
- If k is any scalar in $\mathbb{R}$ and $\vec{u}$ is any vector in V, then $k \cdot \vec{u}$ is a vector in V.

The definition of a vector space $(V, +, \cdot)$

- For any scalar k in $\mathbb{R}$ and any vectors $\vec{u}$ and $\vec{v}$ in V, $k \cdot (\vec{u} + \vec{v}) = k \cdot \vec{u} + k \cdot \vec{v}$.
- For any scalars k and m in $\mathbb{R}$ and any vector $\vec{u}$ in V, $(k + m) \cdot \vec{u} = k \cdot \vec{u} + m \cdot \vec{u}$.
- For any scalars k and m in $\mathbb{R}$ and any vector $\vec{u}$ in V, $k \cdot (m \cdot \vec{u}) = (k\,m) \cdot \vec{u}$.
- For any vector $\vec{u}$ in V, $1 \cdot \vec{u} = \vec{u}$.

The definition of a vector space $(V, +, \cdot)$

Def: A *vector space* over a field F is a set V with two operations $+ : V \times V \to V$ (vector addition) and $\cdot : F \times V \to V$ (scalar multiplications) that satisfy the following properties:

1. V is an abelian group under $+$.
2. $(ab) \cdot v = a \cdot (b \cdot v)$ for all $a, b \in F$ and $v \in V$.
3. $1 \cdot v = v$ for all $v \in V$.
4. $a \cdot (v + w) = a \cdot v + a \cdot w$ for all $a \in F$ and $v, w \in V$.
5. $(a + b) \cdot v = a \cdot v + b \cdot v$ for all $a, b \in F$ and $v \in V$.

A vector space has more structure than an abelian group, but less structure than a ring (only multiplication by scalars, not multiplication of arbitrary pairs of elements of V).

(Ref: Maps Between Vector Spaces - Salil, Harvard)

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Examples

- $V = F^n$
- $V = \mathbb{C}$, $F = \mathbb{R}$
- $V = \mathbb{Z}^n$, $F = \mathbb{Z}_2$
- $V = F[x]$
- $V = F[x]/\langle p(x)\rangle$
- $V = R$ for a ring R containing F.

Def: Let V be a vector space over of F. Vectors $v_1, \ldots, v_n \in V$ are *linearly independent* iff for every $c_1, \ldots, c_n \in F$, if $c_1 v_1 + \cdots + c_n v_n = 0$, then $c_1 = \cdots = c_n = 0$. The vectors $v_1, \ldots, v_n$ form a *basis* for V iff they are linearly independent and $\mathrm{Span}(v_1, \ldots, v_n) = V$, where $\mathrm{Span}(v_1, \ldots, v_n) = \{c_1 v_1 + \cdots + c_n v_n : c_1, \ldots, c_n \in F\}$.

(Ref: Maps Between Vector Spaces - Salil, Harvard)

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Examples of bases

- $(1, 0, 0, \cdots, 0)$, $(0, 1, 0, \ldots, 0)$, $\ldots$, $(0, 0, 0, \ldots, 1)$ is a basis for F^n for every field F.
- **Q:** Is $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$ always a basis for F^3?
- Bases for other examples above?

Def: The *dimension* of a vector space V over F is the size of the largest set of linearly independent vectors in V. (different than Gallian, but we'll show it to be equivalent)

- A measure of "size" that makes sense even for infinite sets.

(Ref: Maps Between Vector Spaces - Salil, Harvard)

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Property of bases

Prop: In a finite-dimensional vector space, every linearly independent set of $\dim(V)$ vectors is a basis. (Later we'll see that all bases have exactly $\dim(V)$ vectors).

Proof: Let $v_1, \ldots, v_k$ be any set of $k = \dim(V)$ linearly independent vectors in V. To show that this is a basis, we need to show that it spans V. Let w be any vector in V. Since $v_1, \ldots, v_k, w$ has more than $\dim(V)$ vectors, this set must be linearly dependent, i.e. there exists constants $c_1, \ldots, c_k, d \in F$, not all zero, such that $c_1 v_1 + \cdots + c_k v_k + dw = 0$. The linear independence of $v_1, \ldots, v_k$ implies that $d \neq 0$. Thus, we can write $w = (c_1/d_1)v_1 + \cdots + (c_k/d_k)v_k$. So every vector in V is in the span of $v_1, \ldots, v_k$.
Q: What are the dimensions of the above examples?

(Ref: Maps Between Vector Spaces - Salil, Harvard)

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Maps Between Vector Spaces

Def (vector-space homomorphisms): Let V and W be two vector spaces over F. A function $f : V \to W$ is a *linear map* iff for every $x, y \in V$ and $c \in F$, we have

1. $f(x + y) = f(x) + f(y)$ (i.e. f is a group homomorphism), and
2. $f(cx) = cf(x)$.

f is an *isomorphism* if f is also a bijection. If there is an isomorphism between V and W, we say that they are *isomorphic* and write $V \cong W$.
Prop: Every n-dimensional vector space V over F is isomorphic to F^n.

Proof: Let $v_1, \dots, v_n$ be a basis for V.
Then an isomorphism from F^n to V is given by:
Corollaries:

- If V is an n-dimensional vector space over a finite field F, then $|V| = |F|^n$.
- If E is a finite field and F is a subfield of E, then $|E| = |F|^n$ for some $n \in \mathbb{N}$. (Q: How does this compare to applying Lagrange's Theorem to E as an additive subgroup of F?)
- if E is a finite field of characteristic p, then $|E| = p^n$ for some $n \in \mathbb{N}$. (Shown on PS7 using Classification of Abelian Groups.)

(Ref: Maps Between Vector Spaces - Salil, Harvard)

Matrices

A linear map $f : F^n \to F^m$ can be described uniquely by an $m \times n$ matrix M with entries from F.

- $M_{ij} = f(e_j)_i$, where $e_j = (000 \cdots 010 \cdots 00)$ has a 1 in the j'th position.
- For $v = (v_1, \ldots, v_n) \in F^n$, $f(v)_i = f(\sum_j v_j e_j)_i = \sum_j v_j f(e_j)_i = \sum_i M_{ij} v_j = (Mv)_i$, where Mv is matrix-vector product.
- Matrix multiplication $\leftrightarrow$ composition of linear maps.
- If $n = m$, then f is an isomorphism $\Leftrightarrow \det(M) \neq 0$.
- Solving $Mv = w$ for v (when given M and $w \in F^m$) is equivalent to solving a linear system with m variables and n unknowns.

Example: $f : \mathbb{Z}_3^3 \to \mathbb{Z}_3^2$ given by $f(v_1, v_2, v_3) = (v_1 + 2v_2, 2v_1 + v_3)$.

(Ref: Maps Between Vector Spaces - Salil, Harvard)

Basis in a vector space V

Definition: A set $S = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\}$ of vectors is called a *basis* for V if

- S is linearly independent
- span $(S) = V$.

In other words, the set $S = \{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\}$ is a basis for V if

- The equation $c_1 \cdot \vec{v_1} + c_2 \cdot \vec{v_2} + \ldots + c_r \cdot \vec{v_r} = \vec{0}$ has only the trivial solution.
- The equation $c_1 \cdot \vec{v_1} + c_2 \cdot \vec{v_2} + \ldots + c_r \cdot \vec{v_r} = \vec{b}$ has a solution for every $\vec{b}$ in V.

Standard bases

Standard bases for the most popular vector spaces

- In $\mathbb{R}^2$: $\{\vec{i}, \vec{j}\}$.
- In $\mathbb{R}^3$: $\{\vec{i}, \vec{j}, \vec{k}\}$.
- In $\mathbb{R}^n$: $\{e_1, e_2, \ldots, e_n\}$.
- In $\mathbf{P}_n$: $1, X, X^2, \ldots, X^n$.
- In $\mathbf{M}_{2\,2}$: all matrices with all entries 0 except for one entry, which is 1. There are 4 such matrices.
- In $\mathbf{M}_{m\,n}$: all matrices with all entries 0 except for one entry, which is 1. There are $m \cdot n$ such matrices.

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Dimension of a vector space

Some vector spaces do not have a *finite* basis.
A vector space has many different bases. However,
Theorem: All bases of a finite dimensional vector space have the **same number** of elements.

Definition: Let V be a finite dimensional vector space. We call *dimension* of V is the number of elements of a basis for V.
We use the notation $\dim(V)$ for the dimension of V.

Example: Counting the elements of the standard basis of each of the popular vectors spaces, we have that:

- $\dim(\mathbb{R}^n) = n$
- $\dim(\mathbf{P}_n) = n + 1$
- $\dim(\mathbf{M}_{m\,n} = m \cdot n$

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The null space, row space and column space of a matrix

Definition: Given an $m \times n$ matrix A, we define:

- The *column space* of $A =$ the subspace of $\mathbb{R}^m$ spanned by the columns of A. Its dimension is called the rank (A).
- The *row space* of $A =$ the subspace of $\mathbb{R}^n$ spanned by its rows.
- The *null space* of $A =$ the solution space of $A \cdot \vec{x} = \vec{0}$.
  It is a subspace of $\mathbb{R}^n$. Its dimension is called the nullity (A).

Theorem: Let A be an $m \times n$ matrix. Then

- rank $(A)+$ nullity $(A) = n$
- dim column space of $A =$ dim row space of A
  $=$ rank $(A) =$ number of leading 1s in the RREF of A.

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Finding Bases

Finding bases for the null space, row space and column space of a matrix
Given an $m \times n$ matrix A

- Reduce the matrix A to the *reduced row echelon form* R.
- Solve the system $R \cdot \vec{x} = \vec{0}$. Find a basis for the solutions space.
  The **same** basis for the solution space of $R \cdot \vec{x} = \vec{0}$ is a basis for the null space of A.
- Consider the non-zero rows of R. They form a basis for the row space of R.
  The **same** basis for the row space of R is a basis for the row space of A.
- Take the columns of R with leading 1s. They form a basis for the column space of R.
  The **corresponding** column vectors in A form a basis for the column space of A.

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Gaussian elimination: reduced row echelon form (RREF)

Example:

$$A = \begin{bmatrix} \boxed{1} & 0 & 0 & 1 \\ 0 & \boxed{1} & 0 & 2 \\ 0 & 0 & \boxed{1} & 3 \end{bmatrix} \text{ is in RREF.}$$

Definition: To be in RREF, a matrix A must satisfy the following 4 properties (if it satisfies just the first 3 properties, it is in REF):

- If a row does not have only zeros, then its first nonzero number is a 1. We call this a *leading* 1.
- The rows that contain only zeros (if there are any) are at the bottom of the matrix.
- In any consecutive rows that do not have only zeros, the leading 1 in the lower row is farther right than the leading 1 in the row above.
- Each column containing leading 1 has zeros in that column.

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Vector Multiplication

Vector Multiplication

Vector Multiplication is slightly complicated that plain Vector Addition. There are a few types of it.

- Scalar into Vector resulting in a vector: e.g. You have a list (a vector) of people's income. Tax rate is 15%. How do you get a list of Tax amounts?
- Vector into Vector resulting in a scalar: e.g. You have different amounts of foreign currencies. You know each ones conversion-to-INR rate. How do you compute total INRs you have?
- Vector into Vector resulting in a vector: e.g. Area of a parallelogram with a right hand rule direction.

Scalar Vector Multiplication

Scalar Vector Multiplication

$$\vec{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\vec{v} \times 2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

Multiply each element of the vector by the scalar

Scalar Vector Multiplication

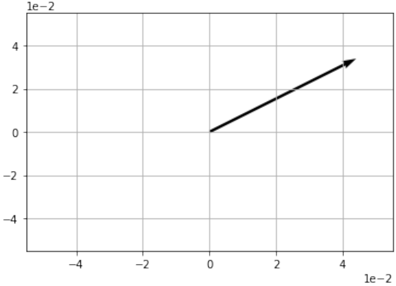
```python
import numpy as np
import matplotlib.pyplot as plt
import math

v = np.array([2,1])

w = 2 * v
print(w)

# Plot w
origin = [0], [0]
plt.grid()
plt.ticklabel_format(style='sci', axis='both', scilimits=(0,0))
plt.quiver(*origin, *w, scale=10)
plt.show()
```

Scalar Vector Multiplication



Scalar Vector Multiplication

$$\vec{b} = \frac{\vec{v}}{2}$$

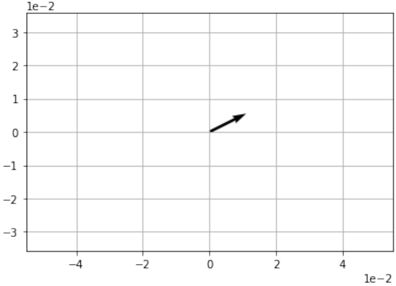
```
b = v / 2
print(b)

# Plot b
origin = [0], [0]
plt.axis('equal')
plt.grid()
plt.ticklabel_format(style='sci', axis='both', scilimits=(0,0))
plt.quiver(*origin, *b, scale=10)
plt.show()

 [1.  0.5]

```

Scalar Vector Multiplication



Dot Product

Vector Vector Multiplication

Dot Product

$$\vec{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \qquad \vec{w} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

$$\vec{v} \cdot \vec{w} = \begin{matrix} (3 \times 2) \\ + \\ (1 \times -4) \end{matrix} = \begin{matrix} 6 \\ + \\ -4 \end{matrix} = 2$$

Multiply the corresponding elements of the vectors and add the results In this case, 3 times 2 is 6, and 1 times -4 is -4; and adding these together gives us our scalar result of 2.

Vector Vector Multiplication

$$\vec{v} \cdot \vec{s} = (v_1 \cdot s_1) + (v_2 \cdot s_2) ... + \ (v_n \cdot s_n)$$

```
import numpy as np

v = np.array([2,1])
s = np.array([-3,2])
d = np.dot(v,s)
print (d)

-4

```

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Vector Vector Multiplication

- Another form: $\vec{v} \cdot \vec{s} = \|\vec{v}\| \|\vec{s}\| \cos(\theta)$
- So for our vectors v (2,1) and s (-3,2), our calculation looks like this:
- $\cos(\theta) = \frac{(2 \cdot -3) + (-3 \cdot 2)}{\sqrt{2^2+1^2} \times \sqrt{-3^2+2^2}}$
- So $\cos(\theta) = -0.496138938357$
- $\theta \approx 119.74$

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Angle Between Two Vectors

- Suppose we have two vectors $\vec{v} = (v, 0)$ lying on X axis and $\vec{w} = (w_1, w_2)$
- $w_1 = ||\vec{w}||cos\theta$, so $\theta = cos^{-1}(\frac{w_1}{||\vec{w}||})$
- Now, dot product is given as $\vec{v} \cdot \vec{w} = v_1.w_1 + 0.w_2 = v_1.w_1$
- Putting value of w_1, eqn becomes $= v_1.||\vec{w}||cos\theta = ||\vec{v}||||\vec{w}||cos\theta$
- Therefore: $cos\theta = \frac{\vec{v} \cdot \vec{w}}{||\vec{v}||||\vec{w}||}$
- Applicable to Higher Dimensions also!!

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Definition

Suppose that $\vec{u}, \vec{v} \in \mathbb{R}^n$. We define the **inner product** or **dot procuct** or $\vec{u}$ and $\vec{v}$ as

$$u \cdot v = u^t v = \sum_{i=1}^{n} u_i v_i.$$

Example

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} = (1)(-1) + (2)(-2) + (3)(1) = -2.$$

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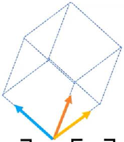
Cross Product

Vector Vector Multiplication

Cross Product (for 3D vectors)

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$



$$\vec{a} \times \vec{b} = \begin{bmatrix} (2\times1)-(3\times2) \\ (3\times3)-(1\times1) \\ (1\times2)-(2\times3) \end{bmatrix} = \begin{bmatrix} -4 \\ 8 \\ -4 \end{bmatrix}$$

Skipping the current row and column, calculate determinant value of remaining sub matrix for that position.

Vector Vector Multiplication

Cross Product

- $\vec{p} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \quad \vec{q} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$
- $$r_1 = p_2 q_3 - p_3 q_2 \tag{1}$$
  $$r_2 = p_3 q_1 - p_1 q_3 \tag{2}$$
  $$r_3 = p_1 q_2 - p_2 q_1 \tag{3}$$
- $\vec{r} = \vec{p} \times \vec{q} = \begin{bmatrix} (3 \cdot -2) - (1 \cdot 2) \\ (1 \cdot 1) - (2 \cdot -2) \\ (2 \cdot 2) - (3 \cdot 1) \end{bmatrix} = \begin{bmatrix} -6 - 2 \\ 1 - -4 \\ 4 - 3 \end{bmatrix} = \begin{bmatrix} -8 \\ 5 \\ 1 \end{bmatrix}$

Vector Vector Multiplication

Cross Product

```python
import numpy as np

p = np.array([2,3,1])
q = np.array([1,2,-2])
r = np.cross(p,q)
print (r)

[-8  5  1]
```

Thanks . . .

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